SPRINT 1 REPORT

Product Backlog, Sprint Review,
Sprint Retrospective, Velocity and Burndown

Wind Turbine Gearbox Predictive Maintenance System
(WT-GPMS)

Scrum project documentation
Sprint duration: 1 week (19th-23rd Oct 2026)

Project Team

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1. Project and Sprint Overview

This report records the work completed on the Wind Turbine Gearbox Predictive Maintenance System (WT-GPMS) as Sprint 1. It includes the Sprint 1 product backlog, completed work, sprint review, sprint retrospective, estimated sprint velocity and a burndown chart. No work outside this completed Sprint 1 scope is included.

Sprint goal: Build a working historical-data pipeline and classical machine-learning anomaly-detection prototype, evaluate the detector and provide a dashboard for reviewing anomaly results.

Duration: 1 week (19th-23rd Oct 2026), represented using five working-day checkpoints.
The burndown is an illustrative reconstruction, not a historical time-tracking export.

2. Product Backlog 1 – Sprint 1 Scope

The following backlog contains the items addressed in the completed Sprint 1. Story points are retrospective estimates assigned for this report, not values exported from a Scrum tracking tool.

ID	Backlog item	Priority	Story points	Outcome
PB01	Collect and understand the five-year SCADA dataset	High	2	Completed
PB02	Validate timestamps, sensor columns and anomaly labels	High	2	Completed
PB03	Explore sensor distributions and anomaly patterns	Medium	2	Completed
PB04	Generate rolling-window statistical features	High	3	Completed
PB05	Prepare model-ready training dataset	High	2	Completed
PB06	Implement Isolation Forest as a comparison experiment	Medium	3	Completed
PB07	Implement and train LOF novelty detector	High	5	Completed
PB08	Apply chronological train/validation/test split	High	2	Completed
PB09	Select threshold using validation data	High	3	Completed
PB10	Evaluate model with classification metrics and ROC-AUC	High	3	Completed
PB11	Perform chronological event-based evaluation	Medium	2	Completed
PB12	Develop prediction/scoring functionality	High	2	Completed
PB13	Build Streamlit monitoring dashboard	High	3	Completed
PB14	Add date filter, anomaly-only view and CSV export	Medium	2	Completed
PB15	Create and run dataset/model automated tests	High	2	Completed
PB16	Prepare README, repository and project documentation	Medium	2	Completed

Estimated completed scope: 40 story points across 16 backlog items. Points are a retrospective estimate for the report and should not be described as automatically recorded project-management data.

3. Sprint 1 Work Completed

Work area	Activities completed	Evidence / deliverables
Dataset preparation	Loaded and inspected the five-year dataset; validated timestamp, sensor fields and binary anomaly label; sorted chronologically.	Validated dataset and validation scripts.
Feature engineering	Calculated 1-hour rolling mean, standard deviation, minimum and maximum for seven sensor channels; derived bearing-minus-oil temperature difference.	Feature dataset with 38 columns including timestamp and label.
Classical ML	Experimented with Isolation Forest; implemented LOF novelty detection with StandardScaler; fitted LOF using normal-labelled training observations.	LOF model artifact and Isolation Forest experiment artifact.
Training and evaluation	Used chronological 70:15:15 split; selected threshold using validation F1; evaluated held-out test period; ran event-based evaluation.	Saved threshold, confusion matrix, classification report, ROC-AUC and event results.
Prediction and dashboard	Implemented prediction/scoring scripts and Streamlit dashboard with date filtering, anomaly-only filtering and CSV export.	Interactive local dashboard.
Testing and repository	Created dataset/model tests; recorded six passing tests; prepared README, requirements and GitHub repository.	Test files, README and repository.

4. Sprint Review

The Sprint Review assessed the working increment against the Sprint 1 goal. The team demonstrated the historical data pipeline, trained LOF model, evaluation outputs and Streamlit dashboard.

Demonstrated increment

- Dataset validation and feature engineering, including rolling 1-hour statistics.
- LOF training using normal-labelled records from the training partition.
- Threshold selection on validation data and evaluation on the chronological test partition.
- Isolation Forest experiment as an additional classical ML approach.
- Prediction/scoring scripts and saved model artifact.
- Streamlit dashboard with date filtering, anomaly-only filtering and CSV export.
- Automated dataset and model tests.

Latest recorded LOF evaluation

Metric	Recorded result
Accuracy	0.87 (87%)
Anomaly precision	0.23 (23%)
Anomaly recall	1.00 (100%)
Anomaly F1-score	0.37
ROC-AUC	1.0
Selected threshold	0.32823885153740934
Confusion matrix	[[32786, 5134], [0, 1500]]

Review findings

- The data preparation, LOF training, evaluation and dashboard workflow was demonstrated as a functioning prototype.
- The model detected all anomaly-labelled records in the displayed test output, but it also produced 5,134 false positives.
- The anomaly precision of 23% indicates that many alerts require human review.
- ROC-AUC of 1.0 is the recorded run result and should be interpreted alongside the confusion matrix and class imbalance.
- The system detects unusual historical sensor behaviour; the available evidence does not establish confirmed failure prediction or advance warning time.

5. Sprint Retrospective

What went well	What could be improved	Action / learning
The work was divided into validation, feature engineering, model training, evaluation and dashboard components.	Dataset paths, feature consistency and model artifacts required careful coordination.	Keep a single documented data schema and feature list shared by training and prediction.
LOF and Isolation Forest provided a useful comparison of classical anomaly-detection approaches.	The LOF result has a high false-positive count despite high recall.	Review threshold selection, alert precision and event-level evaluation before making operational claims.
The dashboard made historical results easier to inspect.	Local dependencies, model files and package imports can make setup difficult on another machine.	Document environment setup, artifact requirements and reliable launch commands.
Automated tests were added and six tests were recorded as passing.	Testing focused mainly on dataset/model basics; full end-to-end integration coverage is limited.	Add repeatable dashboard and pipeline smoke tests and keep test data small.
The project was placed under Git version control with a README.	Large data and model artifacts are not all stored in the repository.	Document how excluded datasets and model artifacts are obtained or regenerated.

Retrospective summary

Sprint 1 established a complete prototype workflow from historical data preparation through LOF-based scoring and dashboard inspection. The most important technical learning was that high recall alone is not sufficient: the false-positive count and anomaly

precision must also be considered. The team also learned to preserve chronological ordering, keep target labels out of model inputs and maintain consistent feature order between training and prediction.

6. Sprint Velocity

Velocity is the sum of story points completed during a sprint. Since the project work was not tracked in a Scrum tool with contemporaneous estimates, the values below are retrospective estimates created to provide a consistent sprint report.

Sprint	Committed points (estimated)	Completed points (estimated)	Completion
Sprint 1	40	40	100%

Estimated Sprint 1 velocity = 40 story points. This is a one-sprint estimate, not a stable team velocity forecast. A reliable average velocity normally requires several completed sprints.

7. Sprint Burndown Chart

The following burndown chart is an illustrative reconstruction for the five working days of Sprint 1 (19th-23rd Oct 2026). It starts with 40 estimated story points and reaches zero by the sprint end. The ideal line decreases evenly; the illustrative actual line represents a plausible completion pattern. It is not derived from dated daily task logs.

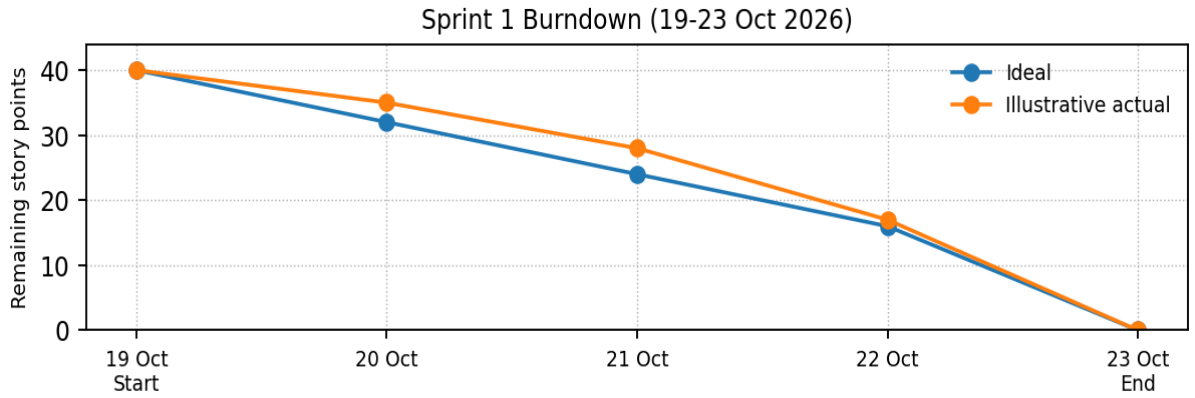


Figure 1. Sprint 1 burndown chart (illustrative remaining story points across five working days).

8. Sprint 1 Definition of Done

• Historical data validation and feature engineering scripts execute successfully.

• The LOF pipeline trains on normal-labelled training records and saves the model artifact, feature list and threshold.

• Evaluation outputs are generated and interpreted with the false-positive limitation clearly stated.

• The dashboard can display historical data and provide the implemented filters and CSV export.

• Dataset and model tests execute; six passing tests were previously recorded.

• README and repository structure provide the basic project setup and usage information.

9. Scope and Evidence Notes

Story points, velocity and burndown checkpoints are retrospective estimates because no dated daily task log was supplied. The technical results are taken from the recorded LOF run. The project should be described as historical anomaly detection, not as a validated gearbox failure forecast.

This report is limited to Sprint 1 work. Story points, velocity and burndown checkpoints are explicitly labelled as retrospective estimates because no dated Scrum board or daily time log was supplied. The technical results are taken from the latest recorded LOF run in the project conversation. The project should be described as historical anomaly detection, not as a validated gearbox failure forecasting or predictive-maintenance deployment system.

End of Sprint 1 Report